

Lab: Pattern Design Workshop — The Pattern-Reader's Mind

Objective

Observe and document your own inference processes while reading and executing a crochet pattern, revealing how you function as an active inference agent.

Materials Needed

- A crochet pattern you have never worked before (intermediate difficulty recommended)
- Hook and yarn appropriate for the pattern
- A journal or notebook
- A timer
- A partner from your crochet circle (for Exercise 3)

Exercise 1: Prediction Journaling (40 minutes)

Before picking up your hook, read through the entire pattern and record your predictions in your journal:

1. **Sketch prediction:** Based only on the text (cover any photos), draw a rough sketch of what you think the finished piece will look like. Include estimated proportions and any texture you imagine.
2. **Difficulty prediction:** Rate each section of the pattern on a 1-5 difficulty scale. Where do you expect to struggle?
3. **Time prediction:** Estimate how long each section will take.
4. **Problem prediction:** Where do you think the pattern might be unclear or where you might make mistakes?

Now uncover the photos (if any). Compare your sketch to the actual finished object image.

Record: How accurate were your predictions? Where did your generative model succeed, and where did it fail? What information were you missing?

Exercise 2: Row-by-Row Observation Log (30 minutes)

Work the first 10 rows (or 10 rounds) of the pattern. After each row, pause and record:

Row	Prediction (what I expected)	Observation (what actually happened)	Match?	Action taken
1				
2				
...				

Look for patterns in your log: - Did your predictions improve over the first few rows as your model calibrated? - Where did the biggest mismatches occur? - What actions did you take when predictions and observations diverged (re-read the pattern, count stitches, frog, ask for help)?

Exercise 3: Agent Comparison (30 minutes)

Work with a partner from your crochet circle. Choose a pattern neither of you has seen before and both read it independently. Then compare:

1. **Mental images:** Describe to each other what you picture the finished piece looking like. Where do your mental images agree? Where do they differ?
2. **Difficulty assessments:** Which parts does each of you find potentially tricky? The differences reveal where your generative models have different strengths.
3. **Abbreviation fluency:** Are there abbreviations one of you knows instantly that the other has to look up? This reveals differences in prior training.
4. **Reading strategy:** Did you read top-to-bottom or skip around? Did you look at photos first or text first? These are different inference strategies.

Discuss: What does the comparison reveal about how individual experience shapes the agent's generative model?

Exercise 4: Designing for Different Agents (20 minutes)

Take a simple pattern you know well (a basic granny square, a simple dishcloth). Rewrite it three ways:

1. **For a complete beginner:** What must you make explicit? What priors can you not assume?
2. **For an intermediate crocheter:** What can you abbreviate? What can you leave to "work as established"?
3. **For an expert:** How minimal can the pattern be while still being unambiguous?

Reflect: How does the target agent's generative model shape what information the pattern must contain?

Discussion Questions for the Circle

1. Have you ever had the experience of a pattern "clicking" after the first few rows — where suddenly the logic becomes clear? What changed in your generative model at that moment?
2. Do you tend to read patterns all the way through before starting, or do you dive in row by row? How does your reading strategy reflect your approach to inference?
3. When you encounter an ambiguous instruction, what is your first response — re-read the pattern, look at the photo, try it and see, or ask someone? What does this say about your preferred source of sensory data?

Wrap-Up

Write a brief reflection (3-5 sentences) comparing your experience as a pattern-reading agent to how you approach other complex instructions in life (recipes, assembly instructions, navigation directions). Do you see the same prediction-observation-update patterns?